

Session 12:

- 1.) Import a CSV file containing monthly sales data for three food delivery apps (Zomato, Swiggy, Uber Eats) into Power BI and create a Waterfall chart to visualize the change in total orders from January to June.

ANSWERS:

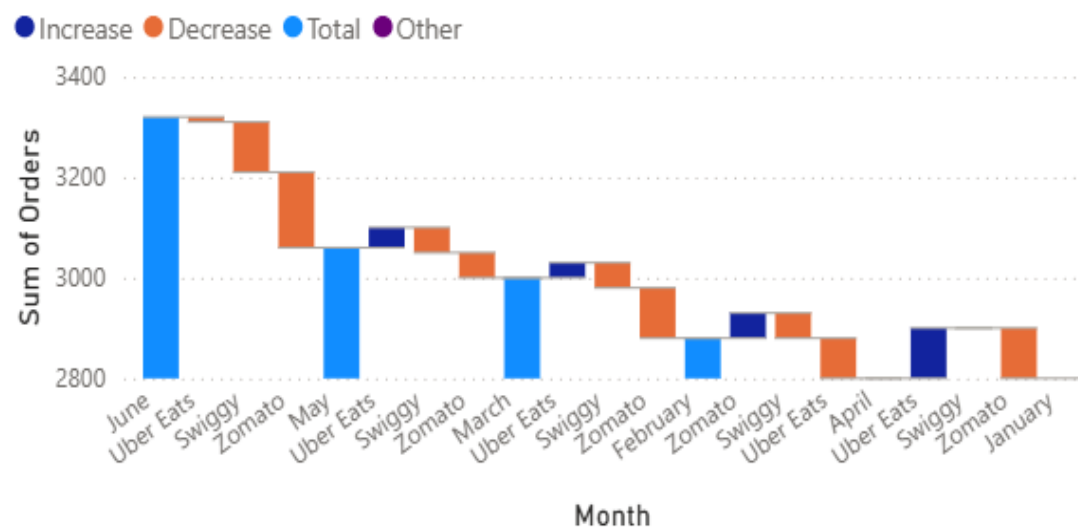
A **Waterfall chart** is created in Power BI after importing the CSV file containing monthly order data for **Zomato, Swiggy, and Uber Eats** from January to June.

The **Month and AppName** fields are placed on the X-axis, and **OrderCount** is used as the value on the Y-axis. This setup helps to visualize how total orders change over the months for each app.

The chart clearly shows the **increase and decrease in orders from January to June**, highlighting how each app contributes to the overall change. Positive values represent growth, while negative values indicate a decline in orders.

Overall, the Waterfall chart provides a clear view of the **cumulative change in total orders**, making it easy to understand trends and performance across different months and apps.

Sum of Orders by Month and AppName



- 2.) Using a dataset of user sign-ups for an IPL fantasy league app, build a Funnel chart in Power BI that shows the number of users at each stage: Visited App, Registered, Created Team, Joined League, Made Payment.

ANSWERS:

A **Funnel chart** is created in Power BI using the dataset of user sign-ups for the IPL fantasy league app.

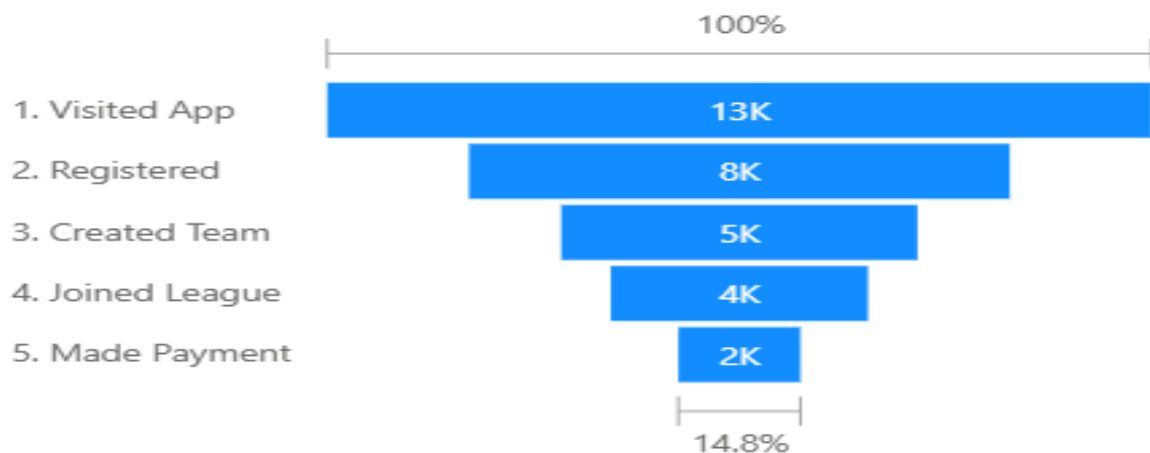
The **Stage** field (Visited App, Registered, Created Team, Joined League, Made Payment) is placed as the category, and **UserCount** is used as the value. This setup helps to represent the number of users at each stage of the process.

The chart clearly shows a **step-by-step decrease in users** from the first stage to the final stage. It highlights how many users move forward and how many drop off at each step.

For example, a large number of users visit the app, but fewer users register, create a team, join a league, and finally make a payment. This makes it easy to identify **where the maximum drop-off is happening**.

Overall, the Funnel chart provides a clear view of the **user conversion process**, helping to understand user behavior and improve the performance of each stage.

Sum of UserCount by Stage



- 3.) Download a sample dataset with city-wise Flipkart order volumes across India and create a Map visual in Power BI to display which cities have the highest and lowest order counts.

Hint: Use the city names as the location field and order volume as the size/colour field in the Map visual.

ANSWERS:

A **Map visual** is created in Power BI after downloading the dataset containing city-wise Flipkart order volumes across India.

The **City** field is placed in the **Location** section, and **OrderCount** is used as the value for **Size** (and optionally Color). This setup helps to plot different cities on the map based on their order volumes.

The map clearly shows **circles on different cities**, where **larger circles represent higher order counts** and smaller circles represent lower order counts. This makes it easy to identify which cities have the highest and lowest number of orders.

For example, cities with bigger bubbles indicate higher demand, while smaller bubbles show lower activity.

Overall, the Map visual provides a clear geographical view of **order distribution across India**, helping to quickly compare performance across different cities.

Sum of OrderCount by City



- 4.) Given a dataset of movies released in 2023 with their genres and box office collections, create a Tree Map in Power BI to show collections by genre, and within each genre, by individual movie.

ANSWERS:

A **Tree Map** is created in Power BI using the dataset of movies released in **2023**, which includes their **genres** and **box office collections**.

In this visual, the **Genre** field is placed as the main category (Group), and **Movie_Title** is added as a sub-category (Details). The **Box_Office_Collection_Millions** field is used as the value to represent the size of each block.

Each rectangle in the Tree Map represents a movie, and its size is based on its box office collection. Movies are grouped under their respective genres, making it easy to compare both **genre-wise performance** and **individual movie contributions**.

The Tree Map clearly shows which genres generated higher collections and which movies performed best within each genre. Larger blocks indicate higher revenue, while smaller blocks represent lower collections.

Overall, this visual helps in understanding the **distribution of box office collections across genres and movies** in a compact and easy-to-read format.

Count of Genre by Movie_Title and Box_Office_Collection_Millions

Avatar: The Way...	Fast X	M3GAN	Missio...	Oppe...
283	704.8			
Barbie	Five Nights at Fr...	181	567.5	957.3
1441.8	291.4			
Cruella 2 Rumors	Guardians of th...	Spider-Man: Acr...	Wonka	
50	845.6	690.5		
Elemental	John Wick: Cha...	The Super Mari...		
496.4	440.1	1362		634.4

- 5.) Use Power BI's Scatter Chart visual to plot the relationship between average daily active users and average session duration for 10 popular Indian apps (e.g., Instagram, WhatsApp, Zomato, Flipkart). Identify any outliers and explain what they might mean.

Hint: Add data labels for at least two points you consider outliers.

ANSWERS:

A **Scatter Chart** is created in Power BI to show the relationship between **average daily active users** and **average session duration** for different apps like Instagram, WhatsApp, Zomato, and Flipkart.

In this visual, **AppName** is placed on the X-axis, **Avg_Daily_Active_Users_Millions** is used on the Y-axis, and **Avg_Session_Duration_Minutes** is represented through the position/size of points. Each dot represents one app, making it easy to compare user engagement across apps.

From the chart, most apps follow a general pattern where higher users are associated with moderate session duration. However, some **outliers** can be identified. For example, **WhatsApp** shows very high daily active users compared to others, indicating its strong user base. On the other hand, **Instagram** (or similar apps) may show higher session duration even with comparatively fewer users, meaning users spend more time per session.

These outliers indicate different user behavior patterns. Apps with high users but lower session time suggest frequent but short usage, while apps with lower users but higher session duration show deeper engagement.

Overall, the Scatter Chart helps in understanding how **user base and engagement time are related**, and highlights apps that behave differently from the general trend.

